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ETHIOPIAN UNIVERSITY ENTRANCE EXAMINATION (EUEE)
PHYSICS EXAMINATION 2008

BOOKLET CODE: 00

NUMBER OF ITEMS:

SUBJECT CODE: 00

TIME ALLOWED: 00

1. A man is standing on a spring scale in an elevator. When the elevator is at rest, the scale reads 560N. When the elevator starts to move, the scale reads 600N. The elevator has

- ☐ A) An upward acceleration
- ☐ B) A constant down ward speed
- ☐ C) A constant upward speed
- ☐ D) A constant upward speed

2. What is the necessary and sufficient condition for conservation of linear momentum?

- ☐ A) The presence of more linear than non linear forces in the system.
- ☐ B) Net force on the system is zero.
- ☐ C) The presence of only non-linear forces.
- ☐ D) Internal forces exceed external forces.

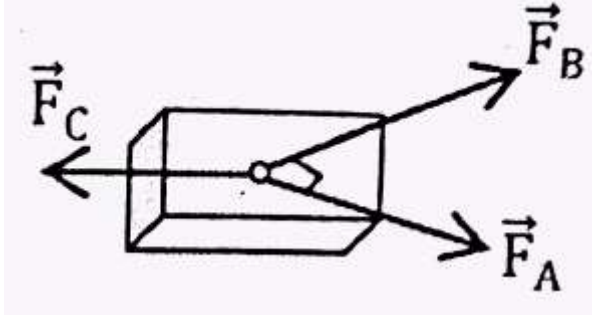
3. Which statement is wrong about units and dimensions?

- ☐ A) Two quantities can have the same units but different dimensions.
- ☐ B) Two quantities can have the same dimensions but different units.
- ☐ C) It is always possible to add two numbers that have the same units.
- ☐ D) There are quantities that have no units and dimension.

4. An airplane flies at speed of 720.0 km/hr at 37° North of west direction. How far does the plane travel to the west in 1 hour?

- ☐ A) 432km
- ☐ B) 720km
- ☐ C) 576km
- ☐ D) 504km

5. Abrehet (A), Bekele(B) and Chaltu(C) are pulling a box horizontally with forces $\vec{F}_A$, $\vec{F}_B$ and $\vec{F}_C$, respectively as shown below, which statement is correct if the box is in equilibrium?



- ☐ A) $\vec{F}_A = \vec{F}_B \times \vec{F}_C$
- ☐ B) $-\vec{F}_A = \vec{F}_B \times \vec{F}_C$
- ☐ C) $\vec{F}_A = \vec{F}_B \cdot \vec{F}_C$
- ☐ D) $\vec{F}_A = \vec{F}_B - \vec{F}_C$

6. Determine the torque in Nm due to a force $\vec{F} = (2\hat{i} + 3\hat{j} + 4\hat{k})\text{N}$ acting at point $\vec{r} = (3\hat{i} + 2\hat{j} + 3\hat{k})\text{m}$ about the origin?

- ☐ A) $6\hat{i} + 6\hat{j} + 12\hat{k}$
- ☐ B) $-6\hat{i} + 6\hat{j} + 12\hat{k}$
- ☐ C) $17\hat{i} - 6\hat{j} + 13\hat{k}$
- ☐ D) $-17\hat{i} + 6\hat{j} + 13\hat{k}$

7. An airplane travelled from Addis Ababa to Mekele, 780km in 45minutes. What was its average speed?

- ☐ A) 0.28m/s
- ☐ B) 1040km/hr
- ☐ C) 17.3m/min
- ☐ D) 585km/hr

8. An example of a body moving with constant speed but still accelerating is

- ☐ A) a body moving on a straight road.
- ☐ B) a body moving on a straight railway track
- ☐ C) a body moving in a circular path
- ☐ D) a body falling in a viscous fluid.

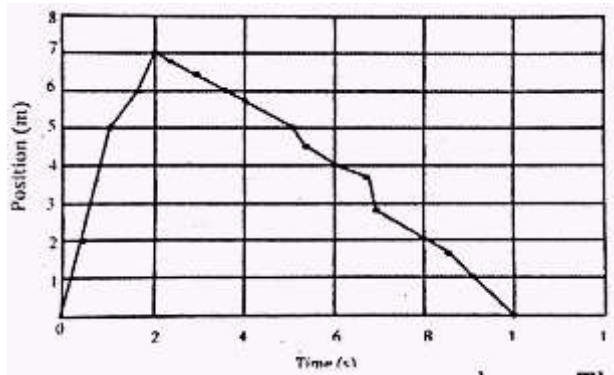
9. A student moves along the boundary of a square field of side 10m in 40sec. if he starts from one end of the corners of the square what will be the magnitude of the displacement of the student at the end of 2 minutes and 20 seconds from his initial position?

- ☐ A) 10m
- ☐ B) $10\sqrt{2}\text{m}$

☐ C) 40m

☐ D) 30m

10. The graph below illustrates the position and time for a cat that runs to catch a rat and then returns with it. The cat caught the rat after 2 seconds. What was the cat's average speed as it returned with rate?



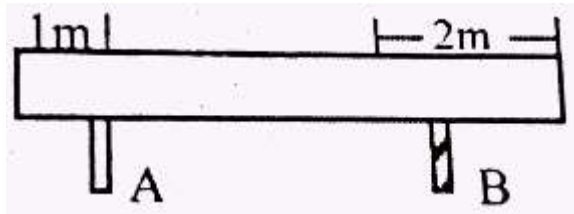
☐ A) its average return speed was 2m/s.

☐ B) its average return speed was 4m/s.

☐ C) its average return speed was 0.7m/s.

☐ D) its average return speed was 0.9m/s.

11. A uniform 6m long and 600N beam, rest on two supports as shown. The forces exerted on the beam by the right support B is closest to:



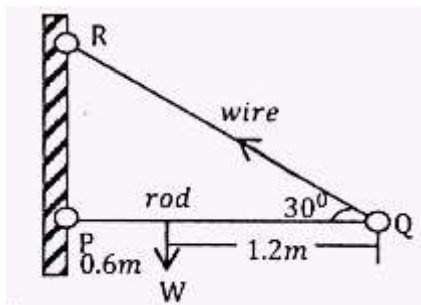
☐ A) 75N

☐ B) 240N

☐ C) 400N

☐ D) 320N

12. A rod PQ is attached at P to a vertical wall, as shown in the figure below. The length of the rod is 1.8m. The weight W of the rod acts at 0.60m from P. The rod is kept horizontal and in equilibrium by a wire attached to Q and to the wall at R. The wire provides a force F on the rod of 50N at 30° to the horizontal. By taking moments about P determine the weight W of the rod.



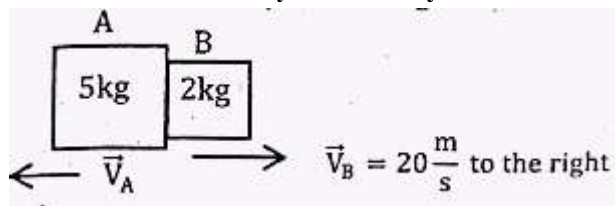
☐ A) 130N

☐ B) 150N

☐ C) 75N

☐ D) 50N

13. A large mass (A) and a small mass (B) are pushing one another apart, as shown in the figure below. The small mass moves away at a velocity of 20m/s to the right. What is the recoil velocity of the large mass?



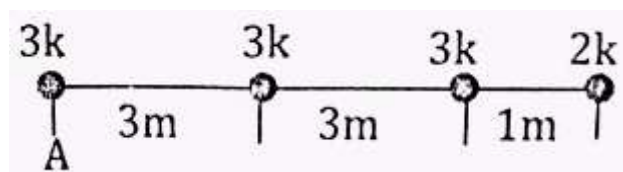
☐ A) -20m/s

☐ B) 8m/s

☐ C) 20m/s

☐ D) -8m/s

14. Find the distance of the center of mass of the system shown in the diagram from point A. The rod is mass less



☐ A) 3.73m

☐ B) 2.09m

☐ C) 2.45m

☐ D) 5.86m

15. The area under a force versus distance graph represents

☐ A) acceleration

☐ B) the change in energy|

☐ C) the change in momentum

☐ D) work done

16. A net external non zero force acts on a particle that is moving along a straight line which one of the following statements is correct?

☐ A) The velocity, but not the kinetic energy of the particle is changing.

☐ B) Both the velocity and kinetic energy of the particle are changing.

☐ C) The kinetic energy but not the velocity of the particle is changing.

☐ D) The momentum is conserved.

17. Which statement is correct about conservative and dissipative force?

☐ A) The work done by conservative forces is path dependent.

☐ B) The work done along a closed path by non-conservative forces is non zero.

☐ C) Gravitational force is an example of non-conservative force.

☐ D) Friction is an example of conservative force.

18. The rotational analogue of mass is

- ☐ A) moment of inertia
- ☐ B) torque
- ☐ C) angular momentum
- ☐ D) angular acceleration

19. A fly wheel is rotating with an angular velocity of 2.0 rad/s acted by an acceleration of $1/\pi \text{ rad/s}^2$. How long will it take to complete three revolutions?

- ☐ A) $0.5\pi\text{sec}$
- ☐ B) πsec
- ☐ C) $2\pi\text{sec}$
- ☐ D) $2\pi\text{sec}^2$

20. If the impulse of a rotating disc is non zero what can you conclude?

- ☐ A) The change in linear momentum is zero.
- ☐ B) The torque acting on object is constant.
- ☐ C) Impulse of the disc equals its moment of inertia.
- ☐ D) The change in angular momentum is non zero.

21. The number of joules of heat energy required to raise the temperature of a body by 1K is

- ☐ A) heat capacity
- ☐ B) specific heat capacity
- ☐ C) expansion coefficient
- ☐ D) calorific value

22. A simple pendulum is made using a long thin metal wire. What happens to the period of the pendulum when the temperature drops?

- ☐ A) increase
- ☐ B) remain the same
- ☐ C) decrease
- ☐ D) period of oscillation has random dependence on temperature.

23. Pressure at the bottom of a liquid column is independent of:

- ☐ A) shape of the column
- ☐ B) the density of the fluid
- ☐ C) the depth of the point
- ☐ D) the acceleration due to gravity.

24. The definition of Young's modulus is given by

- ☐ A) force per unit area
- ☐ B) ratio of strain to stress

- ☐ C) ratio of stress to strain
- ☐ D) force per unit elongation

25. A water pipe has a diameter of 5.0cm in the in let side and 1.5cm in the out let side. Water flows at 0.50m^3 per minute at the in let paint. What is the speed of the water at the out let point?

- ☐ A) 46.99m/s
- ☐ B) 13.33m/s
- ☐ C) 74.1m/s
- ☐ D) 148.1m/s

26. A metal ball has a diameter that is slightly greater than the diameter of a hole that has been cut in to a metal plate. The coefficient of linear expansion for the metal from which the ball is made is greater than that for the metal of the plate. Which one of the following procedures can be used to make the ball pass through the hole?

- ☐ A) raise the temperatures of the ball and the plate by the same amount.
- ☐ B) heat the ball and cool the plate.
- ☐ C) lower the temperatures of the ball and the plate by the same amount.
- ☐ D) there is no way of passing the ball through the hole.

27. A billiard ball of radius 4cm is falling through a liquid of viscosity 0.985 at a velocity of 2.5m/s the drag force on the ball is

- ☐ A) $0.591\pi\text{N}$
- ☐ B) $0.118\pi\text{N}$
- ☐ C) $1.82\pi\text{N}$
- ☐ D) $0.059\pi\text{N}$

28. Which one of the following statements is correct?

- ☐ A) Elastic materials retain their shape after deformation.
- ☐ B) Plastic materials can recover the original size or shape after being distorted.
- ☐ C) Plastic materials change their shape by pressure.
- ☐ D) Elastic materials cannot recover the original size or shape after being distorted.

29. Two bodies that are separately in thermal equilibrium with their body must be in thermal equilibrium with each other. This statement is statement of

- ☐ A) The first law of thermodynamics.
- ☐ B) The zeroth law of thermodynamics.
- ☐ C) The second law of thermodynamics.
- ☐ D) The third law of thermodynamics.

30. A device for transforming heat energy to mechanical energy is referred to as

- ☐ A) transformer
- ☐ B) Refrigerant
- ☐ C) generator

☐ D) engine

31. Three resistors connected in series have individual voltages labelled, Δv_1 , Δv_2 and Δv_3 respectively. Which of the following expresses the value of the voltage Δv_T taken over the three resistors together?

☐ A) $\Delta v_T = \Delta v_1 + \Delta v_2 + \Delta v_3$

☐ B) $\Delta v_T = \Delta v_1 = \Delta v_2 = \Delta v_3$

☐ C) $\Delta v_T = \left(\frac{1}{\Delta v_1} + \frac{1}{\Delta v_2} + \frac{1}{\Delta v_3} \right)$

☐ D) $\Delta v_T = \left(\frac{1}{\Delta v_1} + \frac{1}{\Delta v_2} + \frac{1}{\Delta v_3} \right)^{-1}$

32. Which instrument should ideally have zero resistance?

☐ A) Wheatstone bridge

☐ B) Oscilloscope

☐ C) voltmeter

☐ D) Ammeter

33. What fundamental frequency can a string vibrate at if it is 30cm long and the velocity of traveling waves along the string is 120m/s?

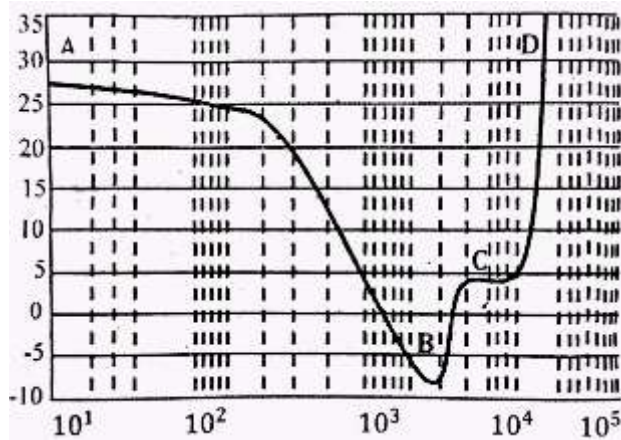
☐ A) 400Hz

☐ B) 600Hz

☐ C) 800Hz

☐ D) 200Hz

34. The hearing curve shown below indicates the vibration of the threshold of hearing with frequency. The ear is most sensitive at the point indicated by



☐ A) A

☐ B) C

☐ C) B

☐ D) D

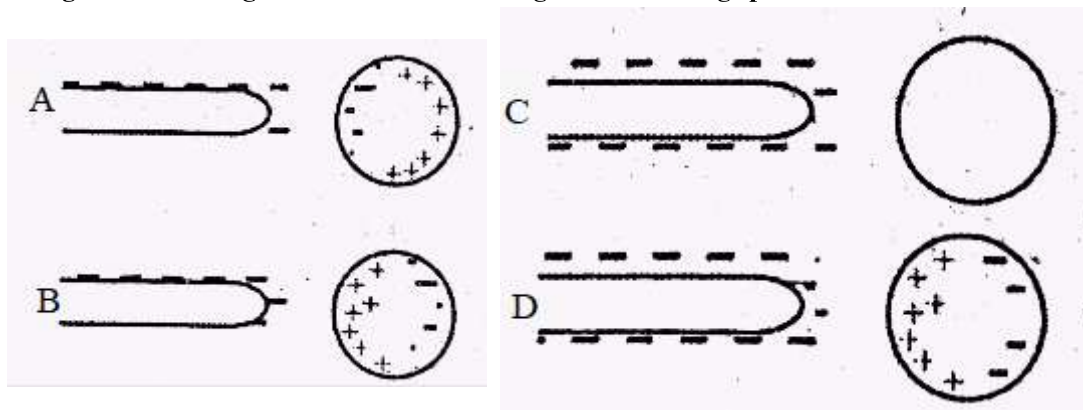
35. A wavelength is

- ☐ A) A line that represents the direction of propagation of light.
- ☐ B) The minimum distance between two points that are in phase.
- ☐ C) The locus of all points of a travelling wave that are in phase.
- ☐ D) The number of waves crossing a given plane in a unit time.

36. The viewed in white light soap bubbles show color because of

- ☐ A) interference
- ☐ B) dispersion
- ☐ C) scattering
- ☐ D) diffraction

37. In the sketch below, which best describes the location of the negative charges (electrons) when a negatively charged rod is brought near but not touching the conducting sphere?

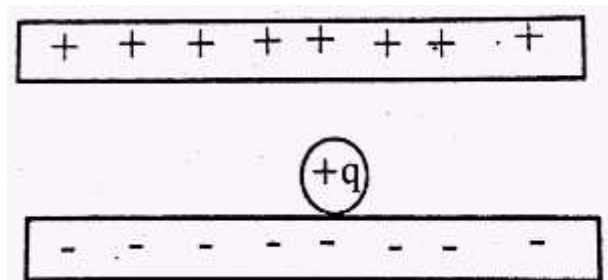


- ☐ A) C
- ☐ B) D
- ☐ C) A
- ☐ D) B

38. What is the potential (V) at a distance of 10m from a charge of +5.0C?

- ☐ A) $24.45 \times 10^9 \text{V}$
- ☐ B) -0.5V
- ☐ C) $4.45 \times 10^9 \text{V}$
- ☐ D) +0.5V

39. A positive charge of $1.0 \times 10^{-4} \text{C}$ is on the negatively charged plate of a parallel plate capacitor as shown below. If the potential(V) on the top plate is +10,000V and that of the bottom plate is -10,000V, how much work (w) is required to move the charge from the bottoms to the top plate?



- ☐ A) 2.0J
- ☐ B) 1.0J
- ☐ C) zero
- ☐ D) 2.0×10^4

40. Which of the following statements is NOT correct about galvanometer?

- ☐ A) it is used to measure small amount of current.
- ☐ B) it has infinite resistance
- ☐ C) it can be used to measure voltage if shunted with large load in series.
- ☐ D) its working principle is the same as that of the electric motor.

41. The activity of a radioactive source falls to one sixteenth of its original value in 32 minutes. What is the half-life in minutes of this decay process?

- ☐ A) 2min
- ☐ B) 8min
- ☐ C) 4min
- ☐ D) 16min

42. Which statements is Not correct about emission from radioactive isotopes?

- ☐ A) Alpha particles are positively charged and less penetrating than beta and gamma.
- ☐ B) Beta particles are negatively charged and more penetrating than alpha particles.
- ☐ C) Alpha emissions are least influenced by magnetic field.
- ☐ D) the most penetrating emissions are gamma particles.

43. The torque experienced by a current carrying loop, in an external, magnetic field is:

- ☐ A) Independent of the number of turns of the loop.
- ☐ B) Independent of the current in the loop.
- ☐ C) Dependent on the orientation of the loop.
- ☐ D) Dependent on magnetic field generated according to Lenz's law.

44. Two long wires of the same length and cross sectional area carry the same amount of current. Which one of the following statements is correct about the magnetic field and the magnetic forces on the wires?

- ☐ A) The magnetic field is zero at the midpoint between the wires when the currents are in the same direction.
- ☐ B) The force is attractive between the wires when they carry current in opposite direction.
- ☐ C) The magnetic fields between the wires reinforce each other when the currents are in the same direction.
- ☐ D) The magnetic field due to one is proportional to the current through the other wire.

45. Self-inductance is a quantity that describes

- ☐ A) The number of turns in a circuit.
- ☐ B) The induction of an electric field by an AC current in a lone coil.
- ☐ C) The induction of a magnetic field by an AC current in a lone coil.
- ☐ D) The steady state current in a coil.

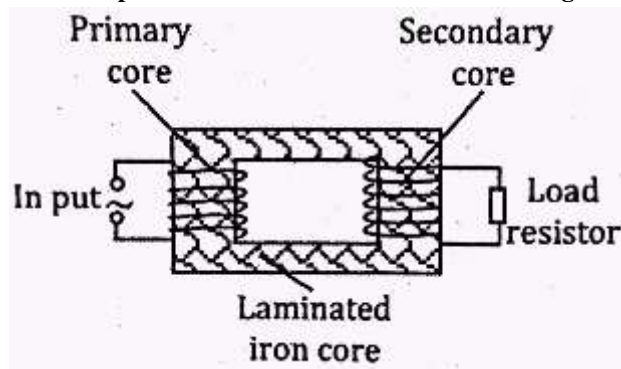
46. Which conversion process forms the bases for the function of an electric generator?

- ☐ A) Alternating current to direct.
- ☐ B) Mechanical energy to electrical.
- ☐ C) Electrical energy to mechanical.
- ☐ D) Low voltage to high voltage or vice versa.

47. What is the change in magnetic flux in a coil of area 5m^2 as its orientation relative to the perpendicular of uniform 3.0T magnetic field changes from 45° to 90° ?

- ☐ A) 0Wb
- ☐ B) -11Wb
- ☐ C) 7.5Wb
- ☐ D) 11Wb

48. A simple transformer is illustrated in the figure below, why is it that the core is laminated?



- ☐ A) To increase the induced current in the primary coil.
- ☐ B) To increase the induced current in the secondary coil.
- ☐ C) To avoid heat exchange between the primary and secondary coils.
- ☐ D) To reduce the loss in power in the core due to eddy currents.

49. A common unit of energy in atomic physics is

- ☐ A) Plank's constant
- ☐ B) Electron volt
- ☐ C) De Broglie wavelength
- ☐ D) Uncertainty principle

50. In photo electric effect, the energy of photoelectrons is proportional to the

- ☐ A) Frequency of light
- ☐ B) Intensity of light
- ☐ C) Speed of light
- ☐ D) Absorbance of the metal

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